



The ARRIVE guidelines 2.0: author checklist

The ARRIVE Essential 10

These items are the basic minimum to include in a manuscript. Without this information, readers and reviewers cannot assess the reliability of the findings.

Item		Recommendation	Section/line number, or reason for not reporting
Study design	1	For each experiment, provide brief details of study design including:	Use cases/Origin of samples
		a. The groups being compared, including control groups. If no control group has been used, the rationale should be stated. b. The experimental unit (e.g. a single animal, litter, or cage of animals).	Use cases/Origin of samples
Sample size	2	a. Specify the exact number of experimental units allocated to each group, and the total number in each experiment. Also indicate the total number of animals used.	Use cases/Origin of samples
		b. Explain how the sample size was decided. Provide details of any <i>a priori</i> sample size calculation, if done.	Use cases/Origin of samples
Inclusion and exclusion criteria	3	a. Describe any criteria used for including and excluding animals (or experimental units) during the experiment, and data points during the analysis. Specify if these criteria were established <i>a priori</i> . If no criteria were set, state this explicitly.	Use cases/Results/ Discarded images
		b. For each experimental group, report any animals, experimental units or data points not included in the analysis and explain why. If there were no exclusions, state so.	Use cases/Results/ Experimental results
		c. For each analysis, report the exact value of <i>n</i> in each experimental group.	Use cases/Origin of samples
Randomisation	4	a. State whether randomisation was used to allocate experimental units to control and treatment groups. If done, provide the method used to generate the randomisation sequence.	Use cases/Origin of samples
		b. Describe the strategy used to minimise potential confounders such as the order of treatments and measurements, or animal/cage location. If confounders were not controlled, state this explicitly.	Use cases/Origin of samples
Blinding	5	Describe who was aware of the group allocation at the different stages of the experiment (during the allocation, the conduct of the experiment, the outcome assessment, and the data analysis).	Use cases/ Histology
Outcome measures	6	a. Clearly define all outcome measures assessed (e.g. cell death, molecular markers, or behavioural changes).	Use cases
		b. For hypothesis-testing studies, specify the primary outcome measure, i.e. the outcome measure that was used to determine the sample size.	NA
Statistical methods	7	a. Provide details of the statistical methods used for each analysis, including software used.	Use cases/Results/ Experimental results
		b. Describe any methods used to assess whether the data met the assumptions of the statistical approach, and what was done if the assumptions were not met.	NA
Experimental animals	8	a. Provide species-appropriate details of the animals used, including species, strain and substrain, sex, age or developmental stage, and, if relevant, weight.	Use cases/Origin of samples
		b. Provide further relevant information on the provenance of animals, health/immune status, genetic modification status, genotype, and any previous procedures.	Use cases/Origin of samples
Experimental procedures	9	For each experimental group, including controls, describe the procedures in enough detail to allow others to replicate them, including:	Use cases/Origin of samples
		a. What was done, how it was done and what was used.	Use cases/Origin of samples
		b. When and how often.	Use cases/Origin of samples
		c. Where (including detail of any acclimatisation periods).	Use cases/Origin of samples
		d. Why (provide rationale for procedures).	Use cases/Origin of samples
Results	10	For each experiment conducted, including independent replications, report:	Use cases/Results/ Experimental results
		a. Summary/descriptive statistics for each experimental group, with a measure of variability where applicable (e.g. mean and SD, or median and range).	NA
		b. If applicable, the effect size with a confidence interval.	